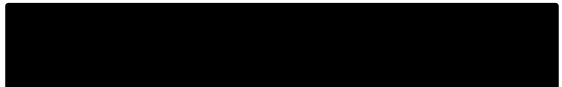
Base conversion (Decimal, binary, octal, hexadecimal)



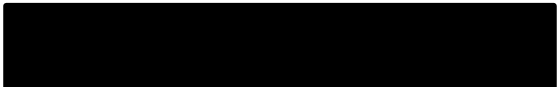
All relevant bases of number systems:

Binary: 0, 1

Octal: 0, 1, 2, 3, 4, 5, 6, 7

Decimal: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

Hexadecimal: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F

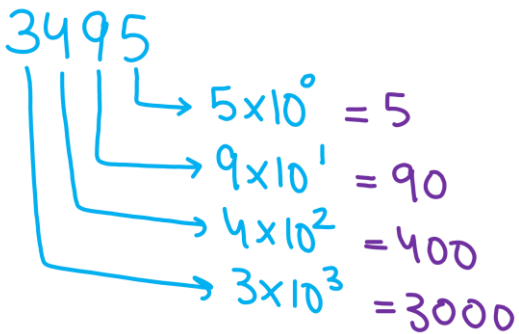


What is base of a number system?

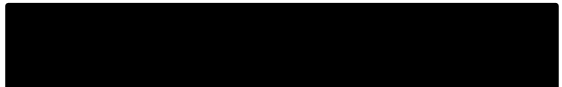
- The number of digits used to create counting

Place value is decided

Example:



How are numbers generated?

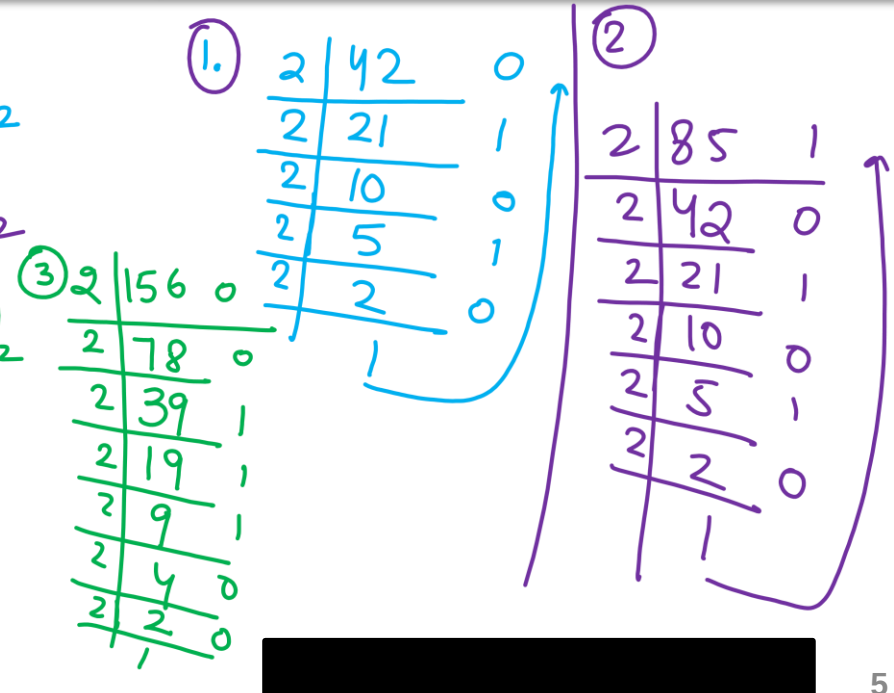


Decimal to binary and vice versa

1) $42_{10} = (101010)_2$

2) $85_{10} = (1010101)_2$

3) $156_{10} = (10011100)_2$



Binary to decimal, last question!

1. $(101010)_2 = (42)_{10}$

$2^5 + 2^3 + 2^1$

$32 + 8 + 2 = 42$

2. $(1010101)_2 = (85)_{10}$

$2^6 + 2^4 + 2^2 + 2^0 = 64 + 16 + 4 + 1 = 85$

3. $(10011100)_2 = (156)_{10}$

$2^7 + 2^4 + 2^3 + 2^2 = 156$

1. Converting the decimal number 241 to the binary number system we get:
दशमलव संख्या 241 को बाइनरी संख्या प्रणाली में परिवर्तित करने पर हमें प्राप्त होता है:

a) 11110000
b) 11110001
c) 10110001
d) 11010001

2. The binary number 11001 in the decimal number system is equal to:
दशमलव संख्या प्रणाली में बाइनरी संख्या 11001 इसके बराबर है:

a) 23
b) 24
c) 25
d) 26

3. **Binary number 11111111 added to 10001000 becomes decimal number:**

बाइनरी संख्या 11111111 को 10001000 में जोड़ने पर दशमलव संख्या बन जाती है:

a) 373

b) 374

c) 391

d) 376

4. The binary number 11111111 divided by 1111 gives _____ in the decimal system.

a) an even number as the answer

b) a composite number as the answer

c) the highest prime number less than 20

d) the second highest prime number less than 20

बाइनरी संख्या 11111111 को 1111 से भाग देने पर दशमलव प्रणाली में _____ प्राप्त होता है।

a) उत्तर के रूप में एक सम संख्या

b) उत्तर के रूप में एक भाज्य संख्या

c) 20 से कम सबसे बड़ी अभाज्य संख्या

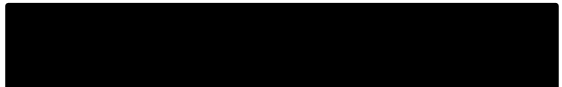
d) 20 से कम दूसरी सबसे बड़ी अभाज्य संख्या

Decimal to Octal and vice versa

1) 59_{10}

2) 120_{10}

3) 345_{10}



Octal to decimal, last page ques:

① 73_8

8^1 8^0

$= 56 + 3 = 59_{10}$

② 170_8

8^2 8^1 8^0

$64 + 56 = 120_{10}$

③ $531_8 =$

8^2 8^1 8^0

$320 + 24 + 1 = 345_{10}$

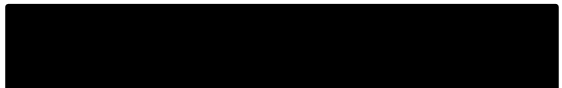
[redacted]

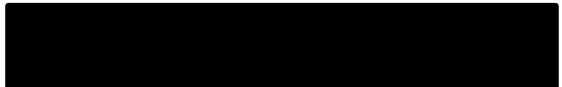
Decimal to Hexadecimal and vice versa

1) 78_{10}

2) 200_{10}

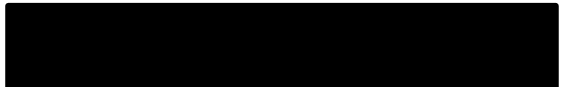
3) 1023_{10}





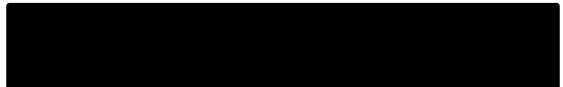
Convert the following numbers to binary, octal and hexadecimal numbers:

1) 25_{10}
2) 58_{10}
3) 127_{10}
4) 255_{10}
5) 345_{10}



Answer key for last question:

Decimal (base-10)	Binary (base-2)	Octal (base-8)	Hex (base-16)
25	11001	31	19
58	111010	72	3A
127	1111111	177	7F
255	11111111	377	FF
345	101011001	531	159



What if I have to convert from binary to octal etc:

